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United States Patent	12403582
Kind Code	B2
Date of Patent	September 02, 2025
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Housing assembly for selectively receiving a part and a method of removing particles from a frictional surface of a pad

Abstract

A housing assembly for selectively receiving a part includes a platform and a pad fixed to the platform. The pad has a frictional surface for maintaining a position of the part relative to the pad when the part is disposed on the frictional surface. The housing assembly includes a door apparatus coupled to the platform. The door apparatus surrounds the pad. The door apparatus is movable to an open position in which the frictional surface of the pad is exposed outside of the door apparatus for receiving the part and a closed position in which the frictional surface of the pad is concealed inside of the door apparatus. A method of removing particles from a frictional surface of a pad before performing a manufacturing process has a door apparatus disposed in a closed position to present a chamber that encloses the pad.

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Appl. No.: 17/476919

Filed: September 16, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220212337 A1	Jul. 07, 2022

Related U.S. Application Data

us-provisional-application US 63133551 20210104

Publication Classification

Int. Cl.: **B25H1/20** (20060101); **B08B3/02** (20060101); B08B5/02 (20060101); B08B15/00 (20060101); B08B15/02 (20060101)

U.S. Cl.:

CPC **B25H1/20** (20130101); **B08B3/02** (20130101); B08B5/02 (20130101); B08B15/00 (20130101); B08B15/02 (20130101)

Field of Classification Search

CPC: B25H (1/20); B08B (3/02); B08B (15/00); B08B (15/02)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to, and the benefit of, U.S. Provisional Patent Application No. 63/133,551 filed on Jan. 4, 2021, which is hereby incorporated by reference in its entirety.

BACKGROUND

(1) During manufacturing processes within a facility, particles may land on various surfaces within the facility. For example, some workstations have a pad supported via a jack, and the pad provides a non-slip surface for resting a part thereon when using that workstation during one or more of the manufacturing processes. However, the pad is open to the environment so that the particles from the manufacturing processes at other workstations may land on the pad before use of that workstation which may affect the characteristics of the non-slip surface.

(2) A manual process of cleaning the pad with water occurs before using the workstation in order to remove the particles from the non-slip surface. This manual process may also require one or more workers to be elevated on a lift to reach the locations of the pad that is to be cleaned.

SUMMARY

(3) Therefore, there is a need for a housing assembly that conceals a pad at a workstation until it is time to use that workstation, which prevents particles from accumulating on the pad when the workstation is not being used. Furthermore, the housing assembly provides a way to clean the pad automatically without requiring one or more workers to be elevated on a lift to reach the locations of the pad that is to be cleaned.

(4) The present disclosure provides a housing assembly for selectively receiving a part. The housing assembly includes a platform and a pad fixed to the platform. The pad has a frictional surface for maintaining a position of the part relative to the pad when the part is disposed on the frictional surface. The housing assembly also includes a door apparatus coupled to the platform. The door apparatus surrounds the pad. The door apparatus is movable to an open position in which the frictional surface of the pad is exposed outside of the door apparatus for receiving the part and a closed position in which the frictional surface of the pad is concealed inside of the door apparatus.

(5) The present disclosure also provides a method of removing particles from a frictional surface of a pad before performing a manufacturing process. A part is placed on the frictional surface for the manufacturing process. A door apparatus is disposed in a closed position to present a chamber that encloses the pad. An image of the frictional surface of the pad is collected while the door apparatus is in the closed position. The collected image of the frictional surface is compared with a reference image of the frictional surface to determine whether particles have been detected on the frictional surface to be removed. A fluid is directed at the frictional surface of the pad to move the particles off of the frictional surface and suspend the particles inside of the chamber when the particles have been detected to be removed from the frictional surface. The suspended particles are vacuumed out of the chamber.

(6) The detailed description and the drawings or figures are supportive and descriptive of the disclosure, but the claim scope of the disclosure is defined solely by the claims. While some of the best modes and other configurations for carrying out the claims have been described in detail, various alternative designs and configurations exist for practicing the disclosure defined in the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic illustration of a facility including workstations to perform manufacturing

processes, and one or more of the workstations include a housing assembly.

(2) FIG. 2 is a schematic fragmentary perspective view of one end of the housing assembly.

(3) FIG. 3 is a schematic perspective view of the housing assembly of one configuration with a door apparatus in the open position.

(4) FIG. 4 is a schematic perspective view of the housing assembly of FIG. 3 with the door apparatus in the closed position.

(5) FIG. 5 is a schematic perspective view of the housing assembly of another configuration with the door apparatus in the open position.

(6) FIG. 6 is a schematic perspective view of the housing assembly of FIG. 5 with the door apparatus in the closed position.

(7) FIG. 7 is a schematic fragmentary enlarged perspective view of an actuator engaging a gear set as also illustrated in FIGS. 5 and 6.

(8) FIG. 8 is a schematic fragmentary perspective view of a casing as also illustrated in FIGS. 5-7.

(9) FIG. 9 is a schematic perspective view of an optical assembly of a cleaning system, with the second door segment in the closed position, and the first door segment removed to show an example position of a camera and a light relative to a frictional surface of a pad.

(10) FIG. 10 is a schematic perspective view of a fluid applicator of the cleaning system and a vacuum of the cleaning system, with the first door segment in the closed position, and the second door segment removed to show an example position of a plurality of outlets of a tube relative to the frictional surface of the pad.

(11) The present disclosure may be extended to modifications and alternative forms, with representative configurations shown by way of example in the drawings and described in detail below. Inventive aspects of the disclosure are not limited to the disclosed configurations. Rather, the present disclosure is intended to cover modifications, equivalents, combinations, and alternatives falling within the scope of the disclosure as defined by the appended claims.

DETAILED DESCRIPTION

(12) Those having ordinary skill in the art will recognize that all directional references (e.g., above, below, upward, up, downward, down, top, bottom, left, right, vertical, horizontal, etc.) are used descriptively for the figures to aid the reader's understanding, and do not represent limitations (for example, to the position, orientation, or use, etc.) on the scope of the disclosure, as defined by the appended claims. Furthermore, the term "substantially" can refer to a slight imprecision or slight variance of a condition, quantity, value, or dimension, etc., some of which that are within manufacturing variance or tolerance ranges.

(13) Referring to the figures, wherein like numerals indicate like or corresponding parts throughout the several views, a facility **10** that manufactures a part **12** or a plurality of parts **12** is generally shown in FIG. 1. The facility **10** may have one or more workstations **14** to perform multiple manufacturing processes on one or more of the parts **12**. As such, a manufacturing process may occur at one of the workstations **14**, and the part **12** may proceed to another one of the workstations **14** for another manufacturing process, and so on until the part **12** is completed. Alternatively, the part **12** may be completed at one of the workstations **14**.

(14) During the manufacturing process at one or more of the workstations **14**, particles **16** may be circulated inside of the facility **10**, some of which may land on components of one or more of the workstations **14**. The particles **16** may enter the facility **10** from the environment or the particles **16** may be produced during the manufacturing processes. The particles **16** may be macroscopic or microscopic, and non-limiting examples of the particles **16** may include debris, foreign object debris (FOD), dust, flakes, grit, etc.

(15) Generally, a housing assembly **18** may be used in the facility **10** to prevent the particles **16** from accumulating on some of the components when that workstation **14** is not being used. More specifically, the housing assembly **18** protects some of the components of the workstations **14** from the circulated particles **16** when that workstation **14** is not being used for the manufacturing

process, as detailed below. The below discussion describes one housing assembly **18**, however, it is to be appreciated that each of the workstations **14** may use one of the housing assemblies **18**.

(16) The housing assembly **18** selectively receives the part **12**. Therefore, when the manufacturing process is to proceed at the workstation **14** with the housing assembly **18**, the housing assembly **18** opens to receive the part **12**. The part **12** may be any suitable configuration, and non-limiting examples may include an airplane part **12** such as a wing, a body component, body join areas, etc.

(17) Turning to FIGS. **2**, **3** and **4**, the housing assembly **18** includes a platform **20** and a pad **22** fixed to the platform **20**. The pad **22** remains stationary on the platform **20**. Generally, the platform **20** supports the pad **22** and the part **12** is selectively disposed on the pad **22** which is discussed further below.

(18) Optionally, the housing assembly **18** may include a jack **24** (see FIG. **4**) that supports the platform **20**. The jack **24** may adjust a height of the platform **20** relative to the ground to accommodate different sizes of the parts **12**. Therefore, adjusting the height of the platform **20** also adjusts the height of the pad **22** relative to the ground. The jack **24** is illustrated in FIG. **4** only, but would be disposed under the platform **20** in FIGS. **2**, **3**, **5**, **6**, **9**, and **10**.

(19) Turning back to the pad **22**, the pad **22** has a frictional surface **26** for maintaining a position of the part **12** relative to the pad **22** when the part **12** is disposed on the frictional surface **26**. That is, the frictional surface **26** provides a non-slip surface for the part **12**. As such, the frictional surface **26** provides enough friction to maintain the position of the part **12** relative to the pad **22** during the manufacturing process. The amount of friction provided via the frictional surface **26** of the pad **22** may be a predetermined threshold which may be based on engineering requirements, government requirements, etc. As one non-limiting example, a high coefficient of friction may be between the frictional surface **26** of the pad **22** and a surface of the part **12** resting against the frictional surface **26**. The frictional surface **26** may be formed of any suitable material(s), and non-limiting examples may include one or more of an elastomer, a polymer, a metal, an alloy, a composite, a ceramic, a foam, etc., and combinations thereof.

(20) The frictional surface **26** of the pad **22** may face away from the jack **24**. In other words, the frictional surface **26** of the pad **22** may face outwardly or upwardly such that the frictional surface **26** may be exposed in certain situations, such as to receive the part **12**, as discussed below.

(21) As best shown in FIGS. **3-6**, the housing assembly **18** also includes a door apparatus **28** coupled to the platform **20**, and the door apparatus **28** surrounds the pad **22**. Generally, the door apparatus **28** is movable relative to the pad **22** to selectively expose the frictional surface **26** of the pad **22**. More specifically, the door apparatus **28** is movable to an open position (see FIGS. **3** and **5**) in which the frictional surface **26** of the pad **22** is exposed outside of the door apparatus **28** for receiving the part **12** and a closed position (see FIGS. **4** and **6**) in which the frictional surface **26** of the pad **22** is concealed inside of the door apparatus **28**. Therefore, when the door apparatus **28** is in the closed position, the frictional surface **26** of the pad **22** is concealed from the particles **16** outside of the door apparatus **28**. Generally, when the workstation **14** is not being used, the door apparatus **28** is in the closed position to prevent the particles **16** from settling on the frictional surface **26** of the pad **22**, which assists in maintaining a cleaner pad when the workstation **14** is not in use as compared to a workstation **14** that does not use the door apparatus **28**.

(22) Continuing with FIGS. **3-6**, in certain configurations, the door apparatus **28** may include a first door segment **30** and a second door segment **32** movable relative to each other between the open position and the closed position. Furthermore, the door apparatus **28** may include a first bellows **34** attached to the first door segment **30** and the platform **20**, and a second bellows **36** attached to the second door segment **32** and the platform **20**. Generally, the first door segment **30**, the first bellows **34**, the second door segment **32**, and the second bellows **36** cooperate in the closed position to present a chamber **38** that encloses the pad **22**. That is, the frictional surface **26** of the pad **22** is protected from the particles **16** outside of the door apparatus **28** when in the closed position.

(23) The first bellows **34** and the second bellows **36** provide some flexibility for the movement of

the first door segment **30** and the second door segment **32** while maintaining a sealed connection between the platform **20** and the first door segment **30** and the second door segment **32** respectively to prevent the particles **16** from entering the chamber **38** when in the closed position. Generally, the first door segment **30** and the second door segment **32** are formed of different materials from the first bellows **34** and the second bellows **36**. For example, the first bellows **34** and the second bellows **36** are each formed of a flexible material to allow movement of the first door segment **30** and the second door segment **32** without binding and while maintaining the sealed connection. For example, the first bellows **34** and the second bellows **36** may accordion together or retract when the door apparatus **28** moves to the open position (see FIGS. **3** and **5**). As another example, the first bellows **34** and the second bellows **36** may accordion out or extend when the door apparatus **28** moves to the closed position (see FIGS. **4** and **6**). Generally, the first bellows **34** and the second bellows **36** are configured similarly to each other. As non-limiting examples, the flexible material may be a rubber, a soft polymer, etc., and combinations thereof.

(24) The first door segment **30** and the second door segment **32** may be formed of a rigid material which is different from the flexible material of the first bellows **34** and the second bellows **36**. That is, the first door segment **30** and the second door segment **32** maintain a shape regardless of whether the door apparatus **28** is in the open position or the closed position (compare FIGS. **3** and **4**). As non-limiting examples, the rigid material may be a hard polymer, a metal, an alloy, etc., and combinations thereof.

(25) Referring to FIGS. **4** and **6**, the first door segment **30** may include a first distal end **40A** and a second distal end **42A** spaced from each other along a first longitudinal axis **44**. Similarly, the second door segment **32** may include a first distal end **40B** and a second distal end **42B** spaced from each other along a second longitudinal axis **46**. Therefore, the first distal end **40A** of the first door segment **30** and the first distal end **40B** of the second door segment **32** are disposed adjacent to each other at one end; and similarly, the second distal end **42A** of the first door segment **30** and the second distal end **42B** of the second door segment **32** are disposed adjacent to each other at the other end. Generally, in certain configurations, the first longitudinal axis **44** and the second longitudinal axis **46** are disposed substantially parallel to each other.

(26) As best shown in FIGS. **2**, **3** and **5**, the first door segment **30** may include a first pivot **48** disposed at the first distal end **40A** and the second distal end **42A** of the first door segment **30** to allow movement of the first door segment **30** relative to the platform **20**. The first pivot **48** may be secured to the platform **20** to support the first door segment **30**. The first pivot **48** may be disposed along the first longitudinal axis **44**.

(27) Furthermore, as best shown in FIGS. **2**, **3**, and **5**, the second door segment **32** may include a second pivot **50** disposed at the first distal end **40B** and the second distal end **42B** of the second door segment **32** to allow movement of the second door segment **32** relative to the platform **20**. The second pivot **50** may be secured to the platform **20** to support the second door segment **32**. The second pivot **50** may be disposed along the second longitudinal axis **46**.

(28) Continuing with FIGS. **2**, **3**, and **5**, the first door segment **30** may include a first side edge **52A** having the first bellows **34** fixed thereto. Therefore, the first door segment **30** and the first bellows **34** cooperate with the platform **20** to provide a first side portion that may selectively protect the pad **22** from the particles **16**. The first door segment **30** may include a second side edge **54A** spaced from the first side edge **52A** of the first door segment **30**. Specifically, the first side edge **52A** of the first door segment **30** is spaced from the second side edge **54A** of the first door segment **30** transverse to the first longitudinal axis **44**. The first side edge **52A** of the first door segment **30** and the second side edge **54A** of the first door segment **30** extends the length of the first door segment **30**. That is, the first side edge **52A** of the first door segment **30** and the second side edge **54A** of the first door segment **30** extends from the first distal end **40A** of the first door segment **30** to the second distal end **42A** of the first door segment **30**. The length of the first door segment **30** is substantially parallel to the first longitudinal axis **44**.

(29) Referring to FIGS. 2, 3, and 5, the second door segment **32** is configured similarly to the first door segment **30**. The second door segment **32** may include a first side edge **52B** having the second bellows **36** fixed thereto. Therefore, the second door segment **32** and the second bellows **36** cooperate with the platform **20** to provide a second side portion that may selectively protect the pad **22** from the particles **16**. The second door segment **32** may include a second side edge **54B** spaced from the first side edge **52B** of the second door segment **32**. Specifically, the first side edge **52B** of the second door segment **32** is spaced from the second side edge **54B** of the second door segment **32** transverse to the second longitudinal axis **46**. The first side edge **52B** of the second door segment **32** and the second side edge **54B** of the second door segment **32** extends the length of the second door segment **32**. That is, the first side edge **52B** of the second door segment **32** and the second side edge **54B** of the second door segment **32** extends from the first distal end **40B** of the second door segment **32** to the second distal end **42B** of the second door segment **32**. The length of the second door segment **32** is substantially parallel to the second longitudinal axis **46**.

(30) When the first side portion and the second side portion come together in the closed position, the frictional surface **26** of the pad **22** is protected from particles **16** outside of the door apparatus **28** (see FIGS. 4 and 6). Therefore, the second side edge **54A** of the first door segment **30** and the second side edge **54B** of the second door segment **32** engage each other when the door apparatus **28** is in the closed position. Optionally, the second side edge **54A** of the first door segment **30** and/or the second side edge **54B** of the second door segment **32** may include a seal **56** or a gasket to further assist in sealing the chamber **38** from entry of the particles **16** when the door apparatus **28** is in the closed position. The optional seal **56** is not shown in all of the figures to illustrate other features, but it is to be appreciated that the seal **56** may be used in any of the configurations herein.

(31) Referring to FIGS. 4 and 6, the door apparatus **28** may include a gear set **58** coupled to the first door segment **30** and the second door segment **32**. Actuation of the gear set **58** causes the door apparatus **28** to move between the open position and the closed position. More specifically, actuation of the gear set **58** causes the first door segment **30** and the second door segment **32** to move between the open position and the closed position.

(32) The gear set **58** is coupled to the first door segment **30** and the second door segment **32** at one end of the door apparatus **28**. For example, the gear set **58** may be coupled to the first distal end **40A**, **40B** of the first door segment **30** and the second door segment **32**. Therefore, the gear set **58** may be supported via the first pivot **48** and the second pivot **50** at the corresponding first distal end **40A**, **40B** of the first door segment **30** and the second door segment **32**. As another example, the gear set **58** may be coupled to the second distal end **42A**, **42B** of the first door segment **30** and the second door segment **32**. Therefore, the gear set **58** may be supported via the first pivot **48** and the second pivot **50** at the corresponding second distal end **42A**, **42B** of the first door segment **30** and the second door segment **32**. It is to be appreciated that the gear set **58** may be disposed at the first distal end **40A**, **40B** of the first door segment **30** and the second door segment **32**, or disposed at the second distal end **42A**, **42B** of the first door segment **30** and the second door segment **32**, or gear sets **58** may be disposed at both of the first distal end **40A**, **40B** and the second distal end **42A**, **42B** of the first door segment **30** and the second door segment **32**.

(33) In certain configurations, the gear set **58** may include a first gear **60** fixed to the first door segment **30**, and a second gear **62** fixed to the second door segment **32**. The first gear **60** may be fixed to the first door segment **30** via the first pivot **48**, and the second gear **62** may be fixed to the second door segment **32** via the second pivot **50**. The first gear **60** and the second gear **62** mesh together such that the first door segment **30** and the second door segment **32** move simultaneously in response to actuation of the gear set **58**. Therefore, the first gear **60** and the second gear **62** may include teeth **64A**, **64B** that mesh with each other.

(34) The first gear **60** may be rotatable about the first longitudinal axis **44** and the second gear **62** may be rotatable about the second longitudinal axis **46**. Therefore, the first door segment **30** is rotatable about the first longitudinal axis **44** and the second door segment **32** is rotatable about the

second longitudinal axis **46**. In certain configurations, the first gear **60** rotates about the first longitudinal axis **44** in an opposition direction from rotation of the second gear **62** about the second longitudinal axis **46** to open and close the door apparatus **28**. Opposite rotation of the first gear **60** and the second gear **62** causes opposite rotation of the first door segment **30** and the second door segment **32**.

(35) Referring to FIGS. **3-6**, the door apparatus **28** may include an actuator **66** coupled to the gear set **58** and selectively activated to move the gear set **58**. Furthermore, the actuator **66** may include an activator **68** coupled to the actuator **66** to selectively activate the actuator **66** to move the gear set **58** which causes movement of the first door segment **30** and the second door segment **32** between the open position and the closed position. The actuator **66** and the activator **68** may be various configurations, and non-limiting examples are shown in FIGS. **3-8** and discussed below.

(36) Referring to FIGS. **3** and **4**, in certain configurations, the actuator **66** may include an electric motor **70**, and the activator **68** may include a switch **72A**. The switch **72A** is in communication with the electric motor **70**, and the gear set **58** is coupled to the electric motor **70**. Therefore, for example, activating the switch **72A** sends a signal to the electric motor **70** to actuate which causes movement of the gear set **58**, and movement of the gear set **58** causes movement of the first door segment **30** and the second door segment **32**. Simply stated, activation of the switch **72A** causes the electric motor **70** to move the door apparatus **28** to the open position and the closed position. In one configuration, the electric motor **70** is connected to the first gear **60**. In other configurations, the electric motor **70** is connected to the second gear **62**. Generally, the switch **72A** is activated via a worker to open and close the door apparatus **28**. The switch **72A** may be any suitable configuration, and non-limiting examples may include a button, a knob, a dial, a toggle, a touch surface, a lever, a motion detector, etc.

(37) Referring to FIGS. **5-8**, in other configurations, the actuator **66** may include a shaft **74** having teeth **64C** that mesh with the gear set **58** such that movement of the shaft **74** in a first direction causes the gear set **58** to rotate which moves the first door segment **30** and the second door segment **32** to the open position, and movement of the shaft **74** in a second direction opposite to the first direction causes the gear set **58** to rotate differently which moves the first door segment **30** and the second door segment **32** to the closed position. More specifically, the teeth **64A** of the first gear **60** or the teeth **64B** of the second gear **62** mesh with the teeth **64C** of the shaft **74**. The part **12** engages a feature (discussed further below) of the shaft **74** which causes the shaft **74** to move in the first direction to open the door apparatus **28**.

(38) The shaft **74** extends to a first end **76** and a second end **78** along a shaft axis **80**. The teeth **64C** of the shaft **74** are disposed axially along the shaft axis **80**. That is, the teeth **64C** of the shaft **74** form a strip **82** of teeth **64C** along the shaft **74**. In certain configurations, the strip **82** of teeth **64C** is disposed substantially parallel to the shaft axis **80**. The strip **82** of teeth **64C** is disposed along an outside **84** of the shaft **74** such that the teeth **64C** are exposed to mesh with the gear set **58**. The shaft axis **80** is disposed transverse to and offset from the first longitudinal axis **44** and the second longitudinal axis **46**. The shaft **74** is movable axially along the shaft axis **80** in the first direction and the second direction. The shaft **74** may be referred to as a probe.

(39) Continuing with the configuration of FIGS. **5-8**, the actuator **66** may include a casing **86** and the shaft **74** is disposed in the casing **86**. The casing **86** may be fixed to the platform **20** and/or the jack **24**. For example, the casing **86** may include a mount **88** to fix the casing **86** to the platform **20** and/or the jack **24**. The shaft **74** is movable relative to the casing **86** along the shaft axis **80**, and the casing **86** may include a slot **90** (see FIG. **8**). The teeth **64C** of the shaft **74** protrude out of the slot **90** such that the teeth **64C** of the shaft **74** face and engage the gear set **58**. That is, the strip **82** of teeth **64C** is movable back and forth within the slot **90** along the shaft axis **80** while engaging the gear set **58**.

(40) Continuing with the configuration of FIGS. **5-8**, the actuator **66** may include a biasing member **92** housed inside the casing **86**. The biasing member **92** continuously biases the shaft **74** in the

second direction. By continuously biasing the shaft **74** in the second direction, the door apparatus **28** is continuously biased to the closed position. The biasing member **92** is disposed between the second end **78** of the shaft **74** and a reaction surface **94** of the casing **86**. The reaction surface **94** of the casing **86** provides a surface for the biasing member **92** to react against to continuously bias the shaft **74** in the second direction. The biasing member **92** may be any suitable configuration, and non-limiting examples of the biasing member **92** may include a spring, a coil spring, a leaf spring, etc.

(41) Optionally, a stop may be coupled to the casing **86** and/or the shaft **74** to prevent the shaft **74** from disconnecting from the casing **86**. If no stop is being used, the biasing member **92** may be sized to bias the shaft **74** to a maximum distance within the casing **86** without causing the shaft **74** to disconnect from the casing **86**.

(42) Continuing with the configuration of FIGS. 5 and 6, the activator **68** may include a cap **96** fixed to the shaft **74**. For example, the cap **96** may be attached to the first end **76** of the shaft **74**. The cap **96** is exposed outside of the casing **86** such that the cap **96** is visible outside of the casing **86**. Furthermore, the shaft **74** and the casing **86** may be disposed outside of the first door segment **30** and the second door segment **32**. As the part **12** moves toward the housing assembly **18**, the part **12** engages the cap **96** which causes actuation of the actuator **66**. As such, the door apparatus **28** moves to the open position as the part **12** engages the cap **96** which moves the shaft **74** in the first direction until the part **12** rests on the frictional surface **26**. In certain configurations, the cap **96** is disposed above the first door segment **30** and the second door segment **32** such that the part **12** engages the cap **96** without the part **12** interfering with the opening of the first door segment **30** and the second door segment **32**.

(43) The housing assembly **18** may also include a cleaning system **98** configured to remove the particles **16** from the frictional surface **26** of the pad **22** when the door apparatus **28** is in the closed position. The cleaning system **98** may operate in multiple functions using various components as detailed below. For example, the cleaning system **98** may operate to determine whether there are particles **16** to be removed from the frictional surface **26** of the pad **22**. As another example, the cleaning system **98** may operate to remove the particles **16** from the frictional surface **26** of the pad **22**. By using the cleaning system **98** discussed below, workers do not have to manually clean the frictional surface **26** of the pad **22**, and thus, do not have to use a lift to reach the frictional surface **26** for cleaning purposes.

(44) Generally, a controller **100** may be in communication with the cleaning system **98**. Instructions may be stored in a memory **102** of the controller **100** and automatically executed via a processor **104** of the controller **100** to provide the respective control functionality. That is, the controller **100** is configured to execute the instructions from the memory **102**, via the processor **104**. For example, the controller **100** may be a host machine or distributed system, e.g., a computer such as a digital computer or microcomputer, and, as the memory **102**, tangible, non-transitory computer-readable memory such as read-only memory (ROM) or flash memory. The controller **100** may also have random access memory (RAM), electrically erasable programmable read-only memory (EEPROM), a high-speed clock, analog-to-digital (A/D) and/or digital-to-analog (D/A) circuitry, and any required input/output circuitry and associated devices, as well as any required signal conditioning and/or signal buffering circuitry. Therefore, the controller **100** may include all software, hardware, memory **102**, algorithms, connections, sensors, etc., necessary to control, for example, the cleaning system **98**. As such, a control method operative to control the cleaning system **98** may be embodied as software or firmware associated with the controller **100**. It is to be appreciated that the controller **100** may also include any device capable of analyzing data from various sensors, comparing data, making the necessary decisions required to control and/or monitor the cleaning system **98** and the door apparatus **28**. The controller **100** may be in communication with the cleaning system **98** and the door apparatus **28** via an electrical connection such as hardwires or wirelessly, etc. Optionally, more than one controller **100** may be utilized. The controller

100 may be referred to as a primary logic controller.

(45) As mentioned above, the cleaning system **98** may operate to remove the particles **16** from the frictional surface **26** of the pad **22**. For example, the cleaning system **98** may suspend the particles **16** within the chamber **38** and then remove the suspended particles **16** from the chamber **38**. Generally, the particles **16** are suspended inside the chamber **38** when the door apparatus **28** is in the closed position. Therefore, before the door apparatus **28** opens to receive the part **12**, the frictional surface **26** of the pad **22** may need cleaning. Once the frictional surface **26** of the pad **22** is cleaned, the door apparatus **28** may be opened, and the details of when the door apparatus **28** opens is discussed further below.

(46) Referring to FIGS. **3-6** and **10**, in certain configurations, the cleaning system **98** may include a fluid applicator **106** configured to direct a fluid at the frictional surface **26** of the pad **22** to move the particles **16** off of the frictional surface **26** and suspend the particles **16** inside of the chamber **38**. As indicated above, the fluid applicator **106** operates when the door apparatus **28** is in the closed position which contains the suspended particles **16** inside the chamber **38** until those particles **16** are removed from the chamber **38** which is discussed further below. Non-limiting examples of the fluid applicator **106** may include a blower, a compressor, a fan, a pump, a sprayer, etc.

(47) In certain configurations, the fluid that is directed at the frictional surface **26** is a gaseous fluid. Generally, the gaseous fluid is air. In other configurations, the fluid that is directed at the frictional surface **26** is a liquid fluid. Generally, the liquid fluid is water. In yet other configurations, the fluid that is directed at the frictional surface **26** is a combination of a gaseous fluid and a liquid fluid such that the mixture may be atomized. It is to be appreciated that any suitable fluid may be used to remove the particles **16** from the frictional surface **26**.

(48) Continuing with FIGS. **3-6** and **10**, generally, the cleaning system **98** may include a duct **108** attached to the first door segment **30** or the second door segment **32**. In one configuration, the duct **108** is attached to the first door segment **30** proximal to the second side edge **54A** of the first door segment **30**. Therefore, when the first door segment **30** is in the closed position, the duct **108** is positioned above the frictional surface **26** of the pad **22**. For example, the duct **108** may be positioned generally centered relative to the frictional surface **26** when the door apparatus **28** is in the closed position. The duct **108** may extend the length of the first door segment **30** and/or the second door segment **32**.

(49) The duct **108** may terminate at a closed end **110**. As one non-limiting example, the closed end **110** of the duct **108** may terminate adjacent to the second distal end **42A**, **42B** of the first door segment **30** or the second door segment **32**. It is to be appreciated that more than one duct **108** may be used in the cleaning system **98**, and thus, more than one duct **108** may be attached to the first door segment **30** or the second door segment **32**, or one or more ducts **108** may be attached to the first door segment **30** and one or more other ducts **108** may be attached to the second door segment **32**.

(50) In certain configurations, the fluid applicator **106** may include the duct **108** attached to the first door segment **30** or the second door segment **32**. The first door segment **30** and the second door segment **32** each include an outer surface **112A**, **112B** that faces away from the pad **22** and an inner surface **114A**, **114B** that faces the pad **22**. Generally, the duct **108** may be attached to the inner surface **114A**, **114B** of the first door segment **30** or the second door segment **32**.

(51) Regardless of the location of the duct **108** being attached to the first door segment **30** or the second door segment **32**, the duct **108** extends outside of the door apparatus **28** to connect to a fluid supply **116**. The fluid supply **116** may provide the gaseous fluid and/or the liquid fluid through the duct **108**. For example, the fluid supply **116** may be pressurized air, pressurized water, etc.

(52) In certain configurations, the fluid applicator **106** may include a valve **118** (see FIGS. **3-6**) to open and close the fluid supply **116**. In other words, the valve **118** may operate to allow a flow of the fluid into the chamber **38** and prevent the flow of the fluid into the chamber **38** depending on

the position of the valve **118**. Therefore, when the valve **118** is opened, the fluid flows into the chamber **38**, and when the valve **118** is closed, the fluid does not flow into the chamber **38**. When using the valve **118**, the controller **100** is in communication with the valve **118**, and the controller **100** determines whether to open the valve **118** or close the valve **118**.

(53) The duct **108** includes an inlet **120** connected to the fluid supply **116** to guide the fluid into the door apparatus **28**. The duct **108** may include a plurality of outlets **122** spaced from each other, and the fluid is directed out of the plurality of outlets **122**. For illustrative purposes, the small dashes or dots shown being expelled out of the outlets **122** in FIG. **10** are to generally represent the fluid. In certain configurations, the plurality of outlets **122** may be spaced from each other axially relative to the first longitudinal axis **44**. In other configurations, the plurality of outlets **122** may be spaced from each other axially relative to the second longitudinal axis **46**. The plurality of outlets **122** face the frictional surface **26** to direct the fluid at the frictional surface **26** of the pad **22** to move the particles **16** off of the frictional surface **26** and suspend the particles **16** inside of the chamber **38**. More specifically, the plurality of outlets **122** face away from the outer surface **112A**, **112B** and the inner surface **114A**, **114B** of the corresponding one of the first door segment **30** and the second door segment **32**. The plurality of outlets **122** is positioned to maximize an area of the frictional surface **26** of the pad **22** that the fluid reaches to remove the particles **16**. The controller **100** is in communication with the fluid applicator **106**, and the controller **100** determines when the fluid applicator **106** operates, which will be discussed further below.

(54) A segment of the duct **108** is disposed inside the door apparatus **28** and another segment of the duct **108** is disposed outside of the door apparatus **28**. Specifically, if using one duct **108**, the duct **108** may extend through the first door segment **30** or the second door segment **32** such that the segment of the duct **108** is disposed inside the first door segment **30** or the second door segment **32**, and the another segment of the duct **108** is disposed outside of the first door segment **30** or the second door segment **32**. The segment of the duct **108** that is disposed inside of the door apparatus **28** includes the plurality of outlets **122**, and the another segment of the duct **108** is disposed outside of the door apparatus **28** includes the inlet **120**.

(55) At least a portion of the duct **108** is flexible to allow movement of the first door segment **30** or the second door segment **32** between the open position and the closed position. For example, the flexible portion of the duct **108** may be disposed outside of the door apparatus **28**. That is, the flexible portion of the duct **108** between the fluid supply **116** and outside of the first door segment **30** or the second door segment **32** is flexible to act as a joint to allow the first door segment **30** or the second door segment **32** to move freely between the open position and the closed position (compare FIGS. **3** and **4**, or FIGS. **5** and **6**).

(56) Referring to FIGS. **3-6** and **10**, as another example, the cleaning system **98** may include a vacuum **124** coupled to the chamber **38**. The vacuum **124** is configured to expel the particles **16** that are suspended inside the chamber **38** due to operation of the fluid applicator **106**. Therefore, after the particles **16** are suspended off of the frictional surface **26** of the pad **22**, the vacuum **124** pulls the fluid and the particles **16** out of the chamber **38**, and thus, away from the frictional surface **26** of the pad **22**. As indicated above, the vacuum **124** operates when the door apparatus **28** is in the closed position. As such, the suspended particles **16** are contained inside the chamber **38** until those particles **16** are removed from the chamber **38** via the vacuum **124**. The controller **100** is in communication with the vacuum **124**, and the controller **100** determines when the vacuum **124** operates, which will be discussed further below.

(57) The vacuum **124** may be supported via the platform **20** and/or the jack **24**. Furthermore, the vacuum **124** may include a conduit **126** attached to the door apparatus **28**, and configured to guide the particles **16** out of the chamber **38**. Therefore, when the vacuum **124** operates, the fluid and the particles **16** are expelled from the chamber **38** via the conduit **126**. The conduit **126** may be disposed through the first door segment **30** or the second door segment **32**. As one non-limiting example, the conduit **126** may be disposed through the second distal end **42A**, **42B** of the first door

segment **30** or the second door segment **32**. When the conduit **126** is disposed through the second distal end **42A**, **42B**, the closed end **110** of the duct **108** may terminate adjacent to the second distal end **42A**, **42B** of the first door segment **30** or the second door segment **32**.

(58) At least a portion of the conduit **126** is flexible to allow movement of the first door segment **30** or the second door segment **32** between the open position and the closed position. For example, the flexible portion of the conduit **126** may be disposed outside of the door apparatus **28**. That is, the flexible portion of the conduit **126** between the door apparatus **28** and the vacuum **124** is flexible to act as a joint to allow the first door segment **30** or the second door segment **32** to move freely between the open position and the closed position. The conduit **126** is flexible similar to the duct **108**, and therefore, the illustration of flexibility of the duct **108** in FIGS. **3-6** is also illustrative of the movement of the conduit **126**.

(59) Furthermore, more than one conduit **126** may be used. As such, more than one conduit **126** may be attached to the first door segment **30** or the second door segment **32**, or one or more conduits **126** may be attached to the first door segment **30** and one or more other conduits **126** may be attached to the second door segment **32**. Each of the conduits **126** may be connected to one vacuum **124** or optionally, more than one vacuum **124** may be used that connects to one or more conduits **126**.

(60) A sensor **128** may be coupled to the vacuum **124** to collect data about an amount of the particles **16** being expelled from the chamber **38**. Therefore, the fluid and the particles **16** being pulled out of the chamber **38** flows past the sensor **128** so that the sensor **128** may collect the data about the amount of the particles **16** being expelled from the chamber **38**. The sensor **128** is in communication with the controller **100**, and the controller **100** determines when to shut off the vacuum **124** based on the amount of the particles **16** being expelled.

(61) The controller **100** monitors the sensor **128** and once the data collected via the sensor **128** indicates that the amount of the particles **16** is within a predetermined range, the controller **100** may signal the vacuum **124** to shut off. Generally, when the amount of the particles **16** is within the predetermined range, this indicates that the frictional surface **26** of the pad **22** is clean. The predetermined range of the amount of the particles **16** may be based on engineering requirements, government requirements, etc. Optionally, the controller **100** may signal the fluid applicator **106** to shut off at the same time as the vacuum **124**. It is to be appreciated that if using more than one vacuum **124**, then more than one sensor **128** may be used and the controller **100** may collect the data from each of the sensors **128** and use that data to determine whether the predetermined range is reached.

(62) As also mentioned above, the cleaning system **98** may operate to determine whether there are particles **16** to be removed from the frictional surface **26** of the pad **22**. For example, an image **130** of the frictional surface **26** may be compiled and analyzed to determine whether the frictional surface **26** is at a predetermined clean threshold. The predetermined clean threshold of the frictional surface **26** may be based on engineering requirements, government requirements, etc. Generally, the image **130** is collected when the door apparatus **28** is in the closed position. It is to be appreciated that more than one image **130** may be collected and analyzed.

(63) Referring to FIGS. **2**, **3** and **9**, in certain configurations, the cleaning system **98** may include an optical assembly **132** configured to compile the image **130** of the frictional surface **26** of the pad **22** when the first door segment **30** and the second door segment **32** are in the closed position. The cleaning system **98** may include a camera **134** attached to the first door segment **30** or the second door segment **32**. In certain configurations, the optical assembly **132** may include the camera **134** attached to the first door segment **30** or the second door segment **32**. Generally, the camera **134** may be attached to the inner surface **114A**, **114B** of the first door segment **30** or the second door segment **32**. In one configuration, the camera **134** is attached to the second door segment **32** proximal to the second side edge **54B** of the second door segment **32**. When the second door segment **32** is in the closed position, the camera **134** is positioned above the frictional surface **26** of

the pad 22. For example, the camera 134 may be positioned generally centered relative to the frictional surface 26 when the door apparatus 28 is in the closed position. Therefore, the camera 134 is positioned to maximize an area of the frictional surface 26 of the pad 22 that the image 130 may collect. It is to be appreciated that more than one camera 134 may be used, and thus, more than one camera 134 may be attached to the first door segment 30 or the second door segment 32, or one or more cameras 134 may be attached to the first door segment 30 and one or more other cameras 134 may be attached to the second door segment 32.

(64) The camera 134 is configured to collect the image 130 of the frictional surface 26 of the pad 22. The optical assembly 132, and thus the camera 134, is in communication with the controller 100, and therefore, the controller 100 determines when to take the image 130 (i.e., determines when to operate the camera 134) and the controller 100 uses the image 130 to determine whether the frictional surface 26 of the pad 22 is clean, as discussed further below.

(65) Since the door apparatus 28 is in the closed position when operating the optical assembly 132, the chamber 38 will be dark. Therefore, it may be desirable to illuminate the chamber 38 when taking the image 130. As such, the optical assembly 132 may include a light 136 configured to illuminate the frictional surface 26 of the pad 22 when the first door segment 30 and the second door segment 32 are in the closed position during operation of the camera 134. Using the light 136 may assist in the quality of the image 130 collected. In certain configurations, the light 136 is attached to the first door segment 30 or the second door segment 32. Generally, the light 136 may be attached to the inner surface 114A, 114B of the first door segment 30 or the second door segment 32. In one configuration, the light 136 is attached to the second door segment 32 proximal to the second side edge 54B of the second door segment 32. When the second door segment 32 is in the closed position, the light 136 is positioned above the frictional surface 26 of the pad 22. For example, the light 136 may be positioned generally centered relative to the frictional surface 26 when the door apparatus 28 is in the closed position. Therefore, the light 136 is positioned to maximize an area of the frictional surface 26 of the pad 22 that the light 136 illuminates. It is to be appreciated that more than one light 136 may be used, and thus, more than one light 136 may be attached to the first door segment 30 or the second door segment 32, or one or more lights 136 may be attached to the first door segment 30 and one or more other lights 136 may be attached to the second door segment 32. The light 136 may be any suitable configuration to illuminate the frictional surface 26, and non-limiting examples may include a light bulb, a light-emitting diode (LED), etc. The controller 100 is in communication with the light 136, and the controller 100 determines when to operate the light 136.

(66) The present disclosure also provides a method of removing the particles 16 from the frictional surface 26 of the pad 22 before performing the manufacturing process. The part 12 is placed on the frictional surface 26 for the manufacturing process. Therefore, the particles 16 are removed from the frictional surface 26 first, and then the door apparatus 28 may be opened to receive the part 12 to perform the manufacturing process when the frictional surface 26 is clean, as detailed below.

(67) The door apparatus 28 is disposed in the closed position to present the chamber 38 that encloses the pad 22. The door apparatus 28 remains in the closed position until it is desired to use that workstation 14. By maintaining the door apparatus 28 in the closed position until that workstation 14 is being used, particles 16 from operating other workstations 14 will not be able to settle on the frictional surface 26 of the pad 22 of the workstations 14 that are not in use.

(68) Prior to opening the first door segment 30 and the second door segment 32, it is desirable to determine whether particles 16 need to be removed from the frictional surface 26 of the pad 22. The controller 100 is activated prior to opening the door apparatus 28, and the controller 100 then analyzes the data collected about the particles 16 and determines whether the frictional surface 26 of the pad 22 is clean. If the controller 100 determines that the frictional surface 26 of the pad 22 is clean, then the door apparatus 28 may be opened to receive the part 12. If the controller 100 determines that the frictional surface 26 of the pad 22 is not clean, then the door apparatus 28 does

not open to receive the part **12**, and a cleaning process occurs.

(69) To activate the controller **100** to start this process, in certain configurations, the activator **68** of the electric motor **70** may be activated. The electric motor **70** will not operate to open the door apparatus **28** until the controller **100** determines that the frictional surface **26** is clean by following the process below.

(70) In other configurations, such as when the activator **68** is disposed on the shaft **74**, a separate activator **138** may be activated which is in communication with the controller **100**. The separate activator **138** is not connected to the activator **68** on the shaft **74**. The separate activator **138** may be a switch **72B**. The switch **72B** of the separate actuator **66** may be any suitable configuration, and non-limiting examples may include a button, a knob, a dial, a toggle, a touch surface, a lever, a motion detector, etc. In this configuration, an indicator **140** may be in communication with the controller **100**, and the indicator **140** is used to signal that the housing assembly **18** is ready to receive the part **12** when the controller **100** determines that the frictional surface **26** is clean by following the process below. The indicator **140** may be any suitable configuration, and non-limiting examples of the indicator **140** may include a visual indicator, a sound indicator, a sensory indicator, etc. When the indicator **140** indicates that the housing assembly **18** is ready to receive the part **12**, the part **12** engages the cap **96** of the shaft **74**, which causes the shaft **74** to move in the first direction and causes the door apparatus **28** to move to the open position.

(71) Particles **16** on a macroscopic level and a microscopic level may be captured via the cleaning system **98**. For example, the image **130** may collect the data about the particles **16** on the macroscopic level. As another example, the sensor **128** of the vacuum **124** may collect the data about the particles **16** on the microscopic level. It is to be appreciated that in other configurations the image **130** may collect data about the particles **16** on the microscopic level, and the sensor **128** of the vacuum **124** may collect data about the particles **16** on the macroscopic level.

(72) The image **130** of the frictional surface **26** of the pad **22** is collected while the door apparatus **28** is in the closed position. In certain configurations, collecting the image **130** may include collecting the image **130** via the camera **134**. Therefore, the camera **134** may be configured to collect images **130** of sufficient quality to identify the particles **16** on the macroscopic level and/or the microscopic level. The controller **100** determines when to activate the camera **134** to take one or more of the images **130**, and sends a signal to the camera **134** to take the images **130**.

(73) Furthermore, the light **136** may be activated inside the chamber **38** to illuminate the frictional surface **26** of the pad **22** to assist the camera **134** in collecting the image **130**. As such, the light **136** may be activated before taking the image **130** of the frictional surface **26** of the pad **22**. The controller **100** determines when to activate the light **136**, and sends a signal to the light **136** to illuminate the chamber **38**. Therefore, the light **136** assists in providing images **130** of sufficient quality to be used via the controller **100** to determine whether the frictional surface **26** of the pad **22** is clean.

(74) The collected image **130** of the frictional surface **26** is compared with a reference image **142** of the frictional surface **26** to determine whether particles **16** have been detected on the frictional surface **26** to be removed. The reference image **142** of the frictional surface **26** may be stored in the memory **102** of the controller **100** and accessed when that workstation **14** is to perform the manufacturing process. The reference image **142** may be an image of a clean frictional surface of the pad **22**.

(75) The controller **100** may analyze and compare the collected image **130** with the reference image **142**. The controller **100** may determine whether the amount of particles **16** on the frictional surface **26** of the pad **22** is at the predetermined clean threshold by comparing the collected image **130** and the reference image **142**. Therefore, if the frictional surface **26** of the pad **22** meets the predetermined clean threshold, then the frictional surface **26** is clean, and the door apparatus **28** may move to the open position to receive the part **12**. If the frictional surface **26** of the pad **22** does not meet the predetermined clean threshold, then the method proceeds to cleaning the frictional

surface **26** of the pad **22**.

(76) To clean the frictional surface **26**, it is desirable to suspend the particles **16** inside the chamber **38** away from the frictional surface **26**. The fluid is directed at the frictional surface **26** of the pad **22** to move the particles **16** off of the frictional surface **26** and suspend the particles **16** inside of the chamber **38** when the particles **16** have been detected to be removed from the frictional surface **26**. In certain configurations, directing the fluid may include directing the fluid via the fluid applicator **106**. Therefore, the controller **100** activates the fluid applicator **106** to inject the fluid into the chamber **38**. If using the valve **118**, the controller **100** signals the valve **118** to open which allows the fluid to flow through the tube and out the outlets **122**.

(77) Next, the suspended particles **16** are vacuumed out of the chamber **38**. In certain configurations, vacuuming the suspended particles **16** may include vacuuming the suspended particles **16** via the vacuum **124**. The particles **16**, with the fluid, are vacuumed out of the chamber **38** via the vacuum **124**. Therefore, the controller **100** activates the vacuum **124** to remove the fluid and the particles **16** from the chamber **38**.

(78) The sensor **128** that is coupled to the vacuum **124** may detect the amount of the particles **16** being removed from the chamber **38**. Therefore, the amount of the particles **16** being vacuumed out of the chamber **38** is sensed, via the sensor **128**. The controller **100** collects the data from the sensor **128**, and the controller **100** determines when the amount of particles **16** reach the predetermined range. Once the predetermined range of the particles **16** is met, the controller **100** may shut off the fluid applicator **106** and the vacuum **124**.

(79) Optionally, the optical assembly **132** may be reactivated after the fluid applicator **106** and the vacuum **124** is shut off. That is, the controller **100** may repeat the operation of taking one or more images **130** and comparing the images **130** to the reference image **142** to confirm that the frictional surface **26** of the pad **22** is clean. Once the controller **100** determines the frictional surface **26** is clean, the door apparatus **28** may be actuated to the open position.

(80) If the controller **100** does not determine that the frictional surface **26** is clean, then the fluid applicator **106** may be reactivated to clean the frictional surface **26**, and the optical assembly **132** may be reactivated to determine whether the frictional surface **26** is clean. If the controller **100** still does not identify that the frictional surface **26** is clean, then a manual process to clean the frictional surface **26** may be necessary.

(81) It is to be appreciated that the order or sequence of performing the method as discussed above is for illustrative purposes and other orders or sequences are within the scope of the present teachings. It is to also be appreciated that the method may include other features not specifically identified in the method discussion above. Furthermore, it is to be appreciated that FIGS. **1**, **2**, **9**, and **10** are general illustrations that may be suitable for any of the configurations discussed herein.

(82) Aspects of the present disclosure have been described in detail with reference to the illustrated configurations. Those skilled in the art will recognize, however, that certain modifications may be made to the disclosed structure and/or methods without departing from the scope of the present disclosure. The disclosure is also not limited to the precise construction and compositions disclosed herein. Modifications apparent from the foregoing descriptions are within the scope of the disclosure as defined by the appended claims. Moreover, the present concepts expressly include combinations and sub-combinations of the preceding elements and features.

Claims

1. A housing assembly for selectively receiving a part; the housing assembly comprising: a platform; a pad fixed to the platform, and the pad has a frictional surface for maintaining a position of the part relative to the pad when the part is disposed on the frictional surface; and a door apparatus coupled to the platform and surrounding the pad, and wherein the door apparatus is movable to an open position in which the frictional surface of the pad is exposed outside of the

door apparatus for receiving the part and a closed position in which the frictional surface of the pad is concealed inside of the door apparatus; and an actuator comprising an axially movable shaft mechanically coupled to the door apparatus and configured to move the door apparatus, the shaft being engaged by the part, wherein: a first axial displacement of the axially movable shaft by engagement with the part causes the door apparatus to move to the open position; and a second axial displacement of the axially movable shaft, in a direction opposite to the first axial displacement, occurs in response to a biasing member and causes the door apparatus to return to the closed position upon disengagement of the part.

2. The housing assembly as set forth in claim 1 wherein the door apparatus includes a first door segment and a second door segment movable relative to each other between the open position and the closed position.

3. The housing assembly as set forth in claim 2 wherein the door apparatus includes a first bellows attached to the first door segment and the platform, and a second bellows attached to the second door segment and the platform.

4. The housing assembly as set forth in claim 2 wherein the door apparatus includes a gear set coupled to the first door segment and the second door segment.

5. The housing assembly as set forth in claim 1 further including a cleaning system configured to remove particles from the frictional surface of the pad when the door apparatus is in the closed position.

6. A housing assembly for selectively receiving a part; the housing assembly comprising: a platform; a pad fixed to the platform, and the pad has a frictional surface for maintaining a position of the part relative to the pad when the part is disposed on the frictional surface; a door apparatus coupled to the platform and surrounding the pad, and wherein the door apparatus is movable to an open position in which the frictional surface of the pad is exposed outside of the door apparatus for receiving the part and a closed position in which the frictional surface of the pad is concealed inside of the door apparatus; a cleaning system configured to remove particles from the frictional surface of the pad when the door apparatus is in the closed position; and wherein: the door apparatus includes a first door segment and a second door segment movable relative to each other between the open position and the closed position; the door apparatus includes a first bellows attached to the first door segment and the platform, and a second bellows attached to the second door segment and the platform; and the first door segment, the first bellows, the second door segment, and the second bellows cooperate in the closed position to present a chamber that encloses the pad.

7. The housing assembly as set forth in claim 6 wherein the cleaning system includes a fluid applicator configured to direct a fluid at the frictional surface of the pad to move the particles off of the frictional surface and suspend the particles inside of the chamber.

8. The housing assembly as set forth in claim 7 wherein the fluid applicator includes a duct attached to the first door segment or the second door segment, and wherein the duct includes a plurality of outlets spaced from each other, and the fluid is directed out of the plurality of outlets.

9. The housing assembly as set forth in claim 7 wherein the cleaning system includes a vacuum coupled to the chamber, and configured to expel the particles that are suspended inside the chamber due to operation of the fluid applicator.

10. The housing assembly as set forth in claim 6 wherein the cleaning system includes an optical assembly configured to compile an image of the frictional surface of the pad when the first door segment and the second door segment are in the closed position.

11. The housing assembly as set forth in claim 10 wherein: the optical assembly includes a camera attached to the first door segment or the second door segment, and the camera is configured to collect the image of the frictional surface of the pad; and the optical assembly includes a light configured to illuminate the frictional surface of the pad when the first door segment and the second door segment are in the closed position during operation of the camera.

12. The housing assembly as set forth in claim 6 wherein: the first door segment includes a first distal end and a second distal end spaced from each other along a first longitudinal axis; the second door segment includes a first distal end and a second distal end spaced from each other along a second longitudinal axis; the first door segment includes a first side edge having the first bellows fixed thereto; the first door segment includes a second side edge spaced from the first side edge of the first door segment; the second door segment includes a first side edge having the second bellows fixed thereto; the second door segment includes a second side edge spaced from the first side edge of the second door segment; the second side edge of the first door segment and the second side edge of the second door segment engage each other when the door apparatus is in the closed position; the cleaning system includes a duct attached to the first door segment proximal to the second side edge of the first door segment; the duct includes a plurality of outlets spaced from each other axially relative to the first longitudinal axis and the plurality of outlets face the frictional surface to direct a fluid at the frictional surface of the pad to move the particles off of the frictional surface and suspend the particles inside of the chamber; and the cleaning system includes a camera attached to the second door segment proximal to the second side edge of the second door segment, and the camera is configured to collect an image of the frictional surface of the pad.

13. A housing assembly for selectively receiving a part, the housing assembly comprising: a platform; a pad fixed to the platform, the pad having a frictional surface for engaging the part; a movable door apparatus surrounding the pad, the movable door apparatus being configured to move between an open position, where the frictional surface is exposed, and a closed position, where the frictional surface is concealed; and an actuator comprising an axially movable shaft mechanically coupled to the moveable door apparatus and configured to move the moveable door apparatus, the shaft being engaged by the part, wherein: a first axial displacement of the axially movable shaft by engagement with the part causes the door apparatus to move to the open position; and a second axial displacement of the axially movable shaft, in a direction opposite to the first axial displacement, occurs in response to a biasing member and causes the moveable door apparatus to return to the closed position upon disengagement of the part.

14. The housing assembly of claim 13, further comprising a cleaning system configured to remove particles from the frictional surface of the pad when the movable door apparatus is in the closed position.

15. The housing assembly of claim 14, wherein the cleaning system includes a fluid applicator configured to direct a fluid at the frictional surface of the pad to dislodge particles.

16. The housing assembly of claim 15, wherein the fluid applicator includes a duct with a plurality of outlets positioned to direct fluid toward the frictional surface.

17. The housing assembly of claim 15, further comprising a vacuum configured to remove fluid and dislodged particles from a chamber when the movable door apparatus is in the closed position.

18. The housing assembly of claim 14, wherein the cleaning system includes an optical assembly configured to capture an image of the frictional surface and detect particles prior to cleaning.

19. The housing assembly of claim 18, wherein the optical assembly includes a light source configured to illuminate the frictional surface of the pad when the movable door apparatus is in the closed position.

20. The housing assembly of claim 18, further comprising a controller configured to analyze the image of the frictional surface and determine whether cleaning is required before opening the movable door apparatus.
